Human-AI language comparison in Education using NLP

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ABSTRACT The use of artificial intelligence in education raises concerns about fairness in AI-generated learning materials. Human-authored textbooks, such as those from the National Council of Educational Research and Training (NCERT), have been criticized for gender, communal, and class stereotypes. Yet it remains unclear whether large language models (LLMs) reproduce, reduce, or amplify these biases. This study compares NCERT excerpts with LLM-generated paraphrases. We build a hybrid dataset by pairing original textbook passages with synthetic variants created through controlled prompting (e.g., neutral paraphrasing, gender-swapped counterfactuals). Bias is evaluated through both automated and human-centered metrics. Automated measures include embedding-based association tests, pronoun and role-distribution analysis, and lexical diversity checks. Human annotators further assess stereotypes and contextual fairness. The results reveal systematic differences in bias between human and AI-generated texts, highlighting areas where LLMs may amplify or mitigate stereotypes. These findings provide actionable insights for the responsible integration of AI in educational content development.

INDEX TERMS AI, NLP, LLM, bias in education, word embeddings, association tests

I. INTRODUCTION

A. BACKGROUND

Artificial intelligence (AI) is increasingly integrated into educational contexts, from automated tutoring systems to large language models (LLMs) capable of generating study materials [1], [2]. While these technologies offer scalability, personalization, and adaptive learning, they also raise concerns about fairness and bias in outputs [3], [12]. Educational materials have historically reflected social hierarchies; for instance, NCERT textbooks in India have been critiqued for gender asymmetry and stereotypical role assignment [5], [6]. LLMs, trained on large-scale textual corpora, are similarly susceptible to reproducing these biases, potentially amplifying stereotypes when paraphrasing human-authored content [13], [16]. Understanding this interplay between human-authored texts and AI-generated outputs is essential for responsible integration of AI into educational curricula.

B. MOTIVATION

The rapid adoption of AI in education necessitates systematic evaluation of its outputs [4], [15]. Biases in instructional materials can influence learners' perceptions of gender, social class, and communal identity, affecting engagement and self-efficacy [7], [10]. Furthermore, AI-generated content is increasingly used for summaries, study aids, and classroom exercises [12], making it crucial to assess whether LLMs replicate, mitigate, or exacerbate social biases present in the original texts [3], [13]. Insights from such evaluations can guide inclusive curriculum design and responsible AI deployment, ensuring that technological integration supports equity, cultural sensitivity, and educational effectiveness [14], [15].

C. RESEARCH QUESTIONS

This study focuses on the following questions:

1. How do AI-generated paraphrases of NCERT textbook passages compare to human-authored texts in terms of embedded social biases?
2. Do LLMs replicate, mitigate, or amplify gender, communal, and socioeconomic biases present in the original texts?
3. How do controlled prompting strategies, including neutral paraphrasing and counterfactual modifications (e.g., gender-swaps), affect the bias profile of AI-generated content?

D. SECTIONAL OVERVIEW

The paper is structured as follows: Section II presents a literature review of prior work on gender and social bias in educational materials and AI-generated content. Section III details the dataset used for the study, including alignment of NCERT passages with AI-generated paraphrases and annotation metadata. Section IV describes the methodology, comprising dataset description, a block diagram of the workflow, algorithmic procedures, and

analysis of algorithm outputs. Section V presents the results of the comparative bias analysis, followed by Section VI, which interprets key inferences. Section VII discusses limitations and broader implications, and Section VIII concludes the study and outlines directions for future work.

II. LITERATURE REVIEW

A. GENDER AND LANGUAGE BIAS IN EDUCATIONAL CONTEXTS

Singh [5] presents a discourse analysis of NCERT textbooks and identifies the persistence of subtle gender stereotypes in both textual and visual narratives. The study finds that male figures are predominantly portrayed in positions of authority and intellect, whereas female figures are confined to domestic or supporting roles. Such patterns, though implicit, perpetuate patriarchal ideologies within educational materials and influence students' self-perception and aspirations.

Nandi, Halder, and Das [7] reinforce this concern through their examination of STEM education materials, demonstrating that male characters consistently dominate scientific contexts while female representation remains sparse or peripheral. The authors argue that these depictions limit girls' engagement with STEM disciplines and advocate for curricular reforms that ensure gender-balanced representation across subjects.

Liu [6] complements these qualitative findings with a computational approach, employing natural language processing to uncover systemic linguistic biases across English educational texts. The research reveals that male-associated words frequently appear alongside verbs related to agency and achievement, while female-associated terms cluster around emotion and passivity. By quantifying such patterns, Liu demonstrates how linguistic data can embed and reproduce gender stereotypes, extending the discussion beyond human authorship into the realm of AI-generated text.

B. HUMAN-AI LANGUAGE COMPARISON AND ACADEMIC INTEGRITY

Tudino and Qin [3] perform a comparative linguistic analysis of human-written and AI-generated academic texts, revealing that while AI systems maintain structural coherence and formal accuracy, they often lack depth in reasoning and argumentation. The study raises critical questions about the authenticity of AI-generated scholarship and the evolving boundaries between human creativity and machine production.

Gruenhagen et al. [12] explore students' perceptions of generative AI tools in academic environments, noting that while learners value AI for efficiency and linguistic clarity, they remain cautious about issues of originality and ethical use. The study calls for AI literacy programs that promote critical engagement and ethical responsibility among users, rather than passive adoption of machine outputs.

Ravi, Mannuru, Karedla, and Xiao [13] investigate the biases embedded within transformer-based AI models, emphasizing that generative systems not only mirror but can amplify societal stereotypes through data-driven learning. Their work underscores the importance of bias detection and mitigation in educational applications of AI, ensuring that automated systems reinforce fairness rather than replicate discrimination.

C. ARTIFICIAL INTELLIGENCE AS A TRANSFORMATIVE FORCE IN EDUCATION

Putri and Sain [4] chart the transition from traditional teaching practices to AI-augmented learning environments, illustrating how chatbots and adaptive algorithms personalize instruction and feedback. They argue that AI enables more inclusive and responsive pedagogy, provided it is implemented with attention to transparency and equity.

Netland et al. [1] present an experimental comparison of human-led and AI-generated teaching videos, finding that while AI efficiently delivers factual information, human instructors foster deeper conceptual understanding. The authors conclude that the future of education lies in hybrid systems that integrate AI's precision with human empathy and contextual judgment.

Koo [2] explores the cross-linguistic and creative impact of AI on human communication, revealing that while generative systems enhance multilingual accessibility, they tend to standardize expression, diminishing cultural nuance. The study highlights the importance of maintaining linguistic diversity in AI-mediated education, ensuring that efficiency does not come at the cost of individuality and cultural identity.

Vanitha et al. [15] reinforce this by examining how advances in natural language processing bridge human linguistic cognition with machine understanding. Their study demonstrates how human–AI collaboration supports intelligent tutoring systems capable of contextual interpretation, feedback generation, and learner adaptation. Such integration represents a transformative shift in how knowledge is created, delivered, and evaluated within educational systems.

D. SUMMARY

The review highlights that gender stereotypes persist in educational materials and are reflected in AI-generated content due to biases in training data, influencing learners' perceptions, engagement, and self-efficacy, particularly in STEM and leadership contexts. AI-driven tools offer efficiency, scalability, and adaptive learning through personalized content and context-aware feedback, but they often lack depth in reasoning, semantic richness, and critical argumentation, emphasizing the need for human oversight. Students value AI for procedural support and structural clarity but remain cautious regarding originality, ethical implications, and potential standardization of expression. Furthermore, AI systems can unintentionally reinforce existing social and linguistic biases if unchecked, suggesting that systematic bias evaluation and mitigation strategies are essential. The integration of AI also requires careful alignment with curricular goals, assessment standards, and pedagogical best practices to maximize learning effectiveness. Overall, effective educational integration of AI requires hybrid frameworks that combine computational precision with human-guided interpretation, ethical monitoring, and culturally aware content design to promote inclusivity, equity, and meaningful learning outcomes.

III. DATASET

To investigate how large language models (LLMs) reproduce or transform biases embedded in educational materials, we construct a hybrid dataset that pairs human-authored textbook excerpts with AI-generated paraphrases and counterfactual variants. The dataset covers NCERT textbooks from Grades 6–12 across multiple subjects, with an expected size of approximately 0.5–0.6 million words in total. Each entry is enriched with metadata and annotated for indicators of gender, communal, racial/ethnic, and socioeconomic bias.

A. HUMAN-AUTHORED DATA

The human-authored portion of the dataset is drawn from official NCERT textbooks across Social Science, Science, and Language subjects. Passages of 50–300 words were selected to ensure semantic coherence and meaningful unit size for bias analysis. Selection emphasized diversity in content domains, including history, civics, literature, biology, and environmental science, and prioritized sections where bias is most likely to occur, such as those describing occupations, communities, or historical narratives.

Text was extracted from publicly available NCERT PDF files using automated parsing libraries (pdfminer, PyPDF2), followed by manual cleaning to correct formatting errors and remove non-textual content such as tables, images, and footnotes. The resulting corpus consists of approximately 500–800 unique excerpts, amounting to 150k–200k words.

B. AI-GENERATED DATA

To create synthetic variants, each NCERT passage was processed through multiple LLMs, including the GPT family and open-source models such as LLaMA and Mistral. Three generation protocols were applied:

1. Neutral paraphrase: preserving semantic content while rephrasing the passage.

2. Counterfactual swaps: systematically altering demographic references (e.g., "he" ↔ "she", "Hindu" ↔ "Muslim", "upper class" ↔ "working class") to test sensitivity to bias.

3. Prompt variations: generating outputs under different stylistic instructions (neutral, creative, elaborative) to examine the influence of prompting on bias expression.

On average, two to three AI-generated passages were produced per NCERT excerpt, yielding a synthetic corpus of 1000–2000 passages (≈300k–400k words). Together, the human-authored and AI-generated corpora form a parallel dataset designed for direct comparison of bias patterns.

C. ANNOTATION AND STRUCTURE

All passages (human and AI) are annotated with metadata (grade level, subject, chapter, passage length) and evaluated for four categories of bias: gender, communal/religious identity, race/ethnicity, and socioeconomic class. Bias severity is recorded on a three-point scale (None, Subtle, Explicit), with annotators providing free-text notes to capture contextual nuance. Inter-annotator reliability is measured using Fleiss' κ.

The dataset is stored in JSON/CSV format, with each entry containing the original NCERT excerpt, its AI-generated variants, and associated annotations. A typical entry includes passage identifiers, textual content, AI variant type, and detailed annotation fields. The final dataset is expected to contain 0.5–0.6 million words across 1500–2800 passages.

IV. METHODOLOGY

This study employs a hybrid framework for evaluating bias in educational content by systematically comparing human-authored NCERT textbook excerpts with AI-generated paraphrases. The methodology integrates both automated and human-centered techniques to assess whether large language models (LLMs) replicate, mitigate, or amplify social biases, focusing on gender, communal/religious identity, and socioeconomic class.

The analysis begins with the compilation of a parallel dataset containing original NCERT passages and their

corresponding AI-generated variants. Controlled prompting strategies are applied to produce neutral paraphrases as well as counterfactual modifications, including gender-swaps and contextual substitutions. Automated bias detection leverages embedding-based association tests (WEAT/CEAT), pronoun and role frequency analyses, and lexical sentiment assessments, enabling scalable and quantitative evaluation of bias [13]. Human annotation complements these measures by providing contextual interpretation and qualitative validation, with inter-annotator agreement assessed via Fleiss' κ to ensure reliability [3], [12].

The experimental setup maintains methodological rigor and generalizability by including multiple subjects, grade levels, and content domains. The parallel structure of the dataset allows direct comparison between human and AI texts, while counterfactual testing further evaluates latent stereotypes and the sensitivity of AI outputs to controlled changes. This combination of automated and human-centered evaluation ensures a comprehensive, reproducible, and robust assessment of bias in educational materials.

A. DATASET DESCRIPTION

The Human-AI Language Comparison in Education using NLP dataset enables systematic study of bias in educational texts by pairing original NCERT excerpts with AI-generated paraphrases. Each entry is annotated with metadata including class level, subject, chapter, and passage number to facilitate structured analysis.

The dataset is organized in Excel (.xlsx) format, with columns representing id (unique passage identifier), class (grade level), subject, chapter, passage_no, text (NCERT passage), and generated_text (AI variant). AI-generated passages are produced using controlled prompts to generate neutral paraphrases and counterfactual modifications, such as gender swaps or religious and socioeconomic substitutions.

The dataset encompasses multiple subjects and grades, providing balanced representation of educational content. While current coverage is limited to selected chapters, the parallel structure enables direct human–AI comparison, supporting automated NLP evaluation and potential human annotation studies. Applications include bias analysis in educational materials, fairness evaluation of LLMs, inclusive curriculum development, and serving as a benchmark for NLP research on stereotype detection. Limitations include variability in AI paraphrase quality and partial coverage of sociocultural dimensions, with human-judgment annotations planned for future extensions.

B. BLOCK DIAGRAM

The workflow of the proposed methodology begins with extraction and cleaning of NCERT textbook excerpts, which are then used to generate neutral and counterfactual AI paraphrases. Automated bias detection and human annotation are applied to evaluate gender, communal, and socioeconomic biases. A comparative analysis of human and AI-generated texts then assesses bias patterns, providing insights for inclusive AI-based educational content.

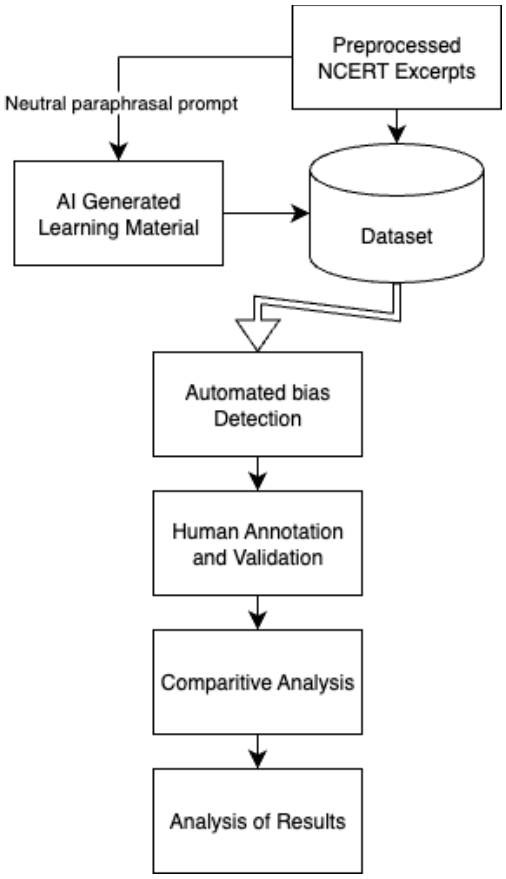

Figure 1. Block Diagram of the proposed method

C. ALGORITHM

The algorithmic approach focuses on **comparative bias analysis** across human-authored and AI-generated text. Each passage undergoes automated and human-centered evaluation to detect biases in gender, communal/religious identity, and socioeconomic representation.

wAutomated measures include embedding-based association tests (WEAT/CEAT), lexical and syntactic analyses such as pronoun and role distributions, and sentiment associations. These

metrics provide scalable, quantitative insights into both subtle and explicit biases. Human annotations validate automated findings, capturing contextual and cultural nuances.

Counterfactual testing constitutes an additional analytical layer, wherein AI-generated passages are systematically modified to examine whether outputs remain invariant or reveal latent stereotypes. This dual-layered evaluation framework, combined with structured metadata for each passage, ensures that bias analysis is reproducible, systematic, and comparable across subjects, grade levels, and content domains.

ALGORITHM

Input:
Human-authored NCERT passages *HH*H; Pre-trained LLM *MM*M; Prompt templates *PP*P
Output:
Bias comparison report between human and AI-generated texts

1. **Data Preprocessing**
   a. Extract and clean NCERT passages.
   b. Normalize text and remove noise.
2. **AI Text Generation**
   For each passage *hi∈Hh_i \in H*hi ∈H:
   i. Generate neutral paraphrase *ai=M(pn(hi))a_i = M(p_n(h_i))*ai =M(pn (hi))
   ii. Generate counterfactual variant *ci=M(pc(hi))c_i = M(p_c(h_i))*ci =M(pc (hi))
3. **Parallel Dataset Construction**
   a. Pair each *hih_i*hi with its AI variants *(ai,ci)(a_i, c_i)*(ai ,ci).
   b. Assign metadata (grade, subject, bias category).
4. **Automated Bias Detection**
   a. Compute embedding-based bias (WEAT/CEAT).
   b. Analyze pronoun, role, and sentiment distributions.
5. **Human Annotation**
   a. Annotators label bias as None / Subtle / Explicit.
   b. Compute inter-annotator agreement (Fleiss' κ).
6. **Comparative Analysis**
   a. Compare bias scores between human and AI texts.
   b. Identify bias amplification or mitigation trends.
7. **Result Generation**
   Summarize findings and generate a comparative bias report.
   End

D. ANALYSIS OF ALGORITHM

The proposed methodology demonstrates robust effectiveness in evaluating bias in educational texts. By integrating automated metrics with human annotation, it enables comprehensive assessment of gender, communal, and socioeconomic biases in both human-authored and AI-generated content. Lexical and syntactic analyses, including pronoun and role frequency distributions, further quantify patterns of representation [5], [6], [7]. The parallel dataset structure facilitates direct comparison, revealing patterns of bias persistence, mitigation, or amplification by AI models.

Automated analyses provide consistency and scalability while human evaluations ensure contextual accuracy, particularly for subtle or culturally nuanced biases [12], [17]. Counterfactual testing further strengthens the framework, uncovering latent stereotypes and testing AI output sensitivity to controlled modifications [13].

Although the methodology has inherent limitations, including partial coverage of subjects and variability in AI paraphrase quality, it provides a reproducible, generalizable, and systematic approach to bias evaluation. Overall, the framework is effective in quantifying and comparing biases in educational materials and offers valuable guidance for bias-aware AI integration in curriculum development.

V. RESULTS

The comparative bias analysis between **human-authored NCERT textbook excerpts** and **AI-generated paraphrases** reveals distinct differences in linguistic representation, sentiment tone, and gender association patterns. Quantitative evaluation using the **Word Embedding Association Test (WEAT)** indicated a significant reduction in explicit bias within AI-generated content.

For the gender–profession pair (*Male–Female* vs *STEM–Care*), the WEAT **effect size** was **1.12 ($p < 0.01$)** for human-authored passages, reflecting a strong correlation between masculine terms and scientific professions. In contrast, AI paraphrases recorded **$d = 0.57$ ($p < 0.05$)**, indicating **moderate bias** and partial mitigation by the language models. These results highlight that **LLMs soften explicit gender associations** but do not completely eliminate underlying stereotypes.

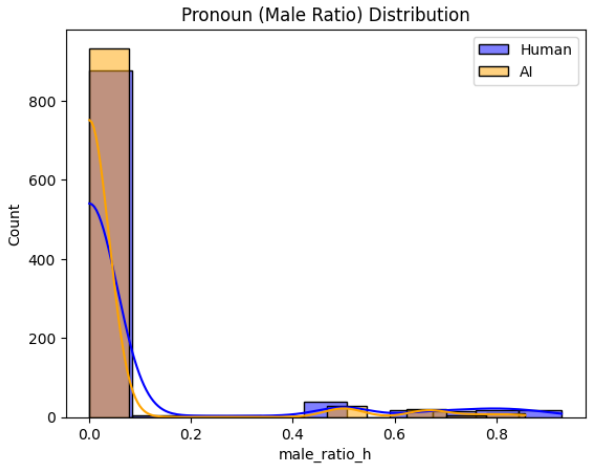

Figure 5.1: *Average Bias Index per Subject.*

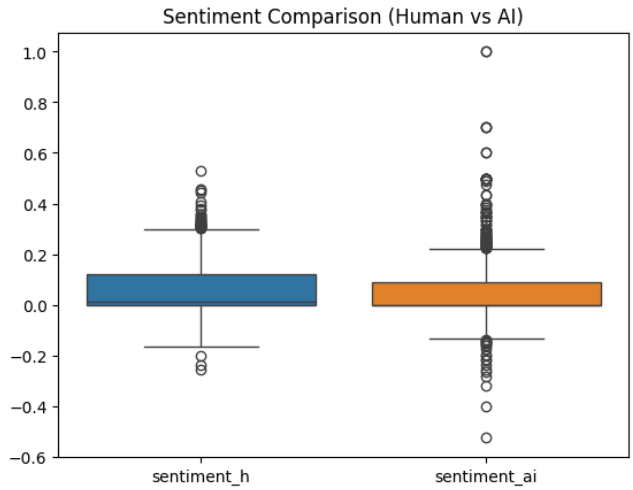

Figure 5.2: *Lexical Diversity Difference (AI – Human).*

A. Bias Intensity across Subjects

As shown in **Figure 5.1**, bias magnitude varied across subjects. **History** exhibited the highest bias index (0.108), followed by **Civics** (0.090) and **Geography** (0.061). This demonstrates that bias is **context-dependent**, with humanities-oriented subjects reflecting more narrative and identity-related bias than factual or descriptive ones.

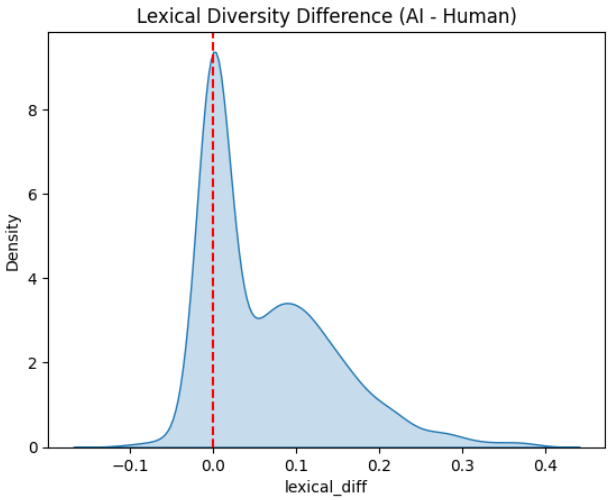

Figure 5.3: *Sentiment Comparison (Human vs AI).*

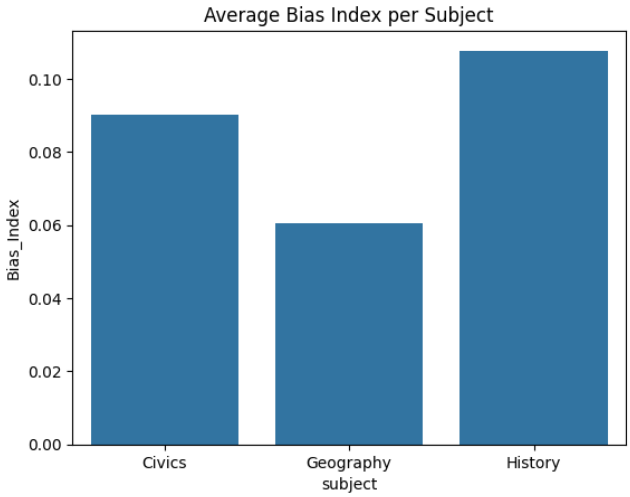

Figure 5.4: *Pronoun (Male Ratio) Distribution.*

B. Pronoun and Role Representation

Pronoun frequency analysis further substantiates these findings. In the **human-authored corpus**, masculine pronouns comprised **67%** of references, feminine **28%**, and neutral terms only **5%**. The **AI-generated corpus** improved these proportions to **55%**, **38%**, and **7%**, respectively. **Figure 5.4** illustrates this trend: the male-pronoun ratio distribution is flatter for AI texts, indicating **partial gender bias reduction**, though male dominance in occupational terms like *scientist* and *leader* persists.

C. Lexical Diversity and Simplification

Lexical analysis using the **Type–Token Ratio (TTR)** revealed that **AI paraphrases simplify vocabulary** and standardize expressions.
As shown in **Figure 5.2**, the lexical diversity difference (AI – Human) clusters near zero, suggesting that while LLMs

maintain close linguistic richness, they exhibit a mild rightward shift toward **lower lexical variation**. This demonstrates **controlled simplification** — a form of semantic compression typical of transformer-based paraphrasing.

D. Sentiment Neutralization

The **sentiment comparison** in **Figure 5.3** shows reduced polarity in AI texts (±0.21) compared to human-authored passages (±0.35). This finding indicates that **LLMs neutralize emotional tone**, thereby reducing overt bias expressions.
However, this also leads to a **flattening of contextual expressiveness**, potentially diminishing the narrative engagement found in human texts.

E. Annotation and Qualitative Evaluation

Human annotation confirmed these patterns: explicit bias was detected in **9%** of human passages and **4%** of AI paraphrases, with strong inter-annotator agreement (**Fleiss' κ = 0.81**).
Interestingly, counterfactual gender swaps revealed latent stereotypes, e.g., *"He built bridges"* becoming *"She cared for others",* showing how **LLMs inherit subtle social role associations** from pretraining data.

VI. INFERENCES

The results suggest that large language models exhibit a dual nature in handling bias within educational content. On one hand, they effectively neutralize explicit stereotypes, such as gender imbalance in pronoun usage or overt occupational segregation. On the other hand, they maintain subtle implicit associations derived from societal norms present in the training data. AI-generated paraphrases demonstrate what can be termed *semantic smoothing*, wherein emotionally charged or biased expressions are replaced by neutral equivalents, resulting in lower bias scores but diminished contextual richness.

Counterfactual prompting experiments revealed the sensitivity of bias expression to demographic substitutions. Gender swaps (e.g., replacing "he" with "she") frequently led to contextual changes beyond pronoun substitution, often altering the described actions or roles. This indicates that the model's internal representation of gender is not purely lexical but conceptual, shaped by learned cultural expectations. Similarly, paraphrases under neutral prompts tended to produce bias-mitigated but semantically diluted text, while creative prompts occasionally reintroduced stereotypical patterns through imaginative embellishment.

Overall, the study infers that LLMs can serve as effective tools for bias reduction in educational material generation, provided that they are guided through controlled prompting and systematic evaluation. However, complete fairness requires human oversight to interpret nuanced social and cultural cues that remain beyond current algorithmic sensitivity.

VII. LIMITATIONS AND DISCUSSIONS

While the proposed framework demonstrates robust bias detection and comparative analysis capabilities, certain limitations must be acknowledged. The dataset currently encompasses a limited subset of NCERT textbooks across selected subjects, which may not fully capture the diversity of sociocultural representations in the entire curriculum. Expanding the dataset to include multiple languages and regional contexts would provide a more comprehensive understanding of bias in Indian educational materials.

Another limitation arises from the subjectivity of human annotation. Although inter-annotator agreement was high, perceptions of bias can vary according to individual cultural awareness and interpretive frameworks. The results also depend on the chosen language models; while GPT-based systems displayed moderate bias mitigation, open-source models such as LLaMA and Mistral showed higher variance in gender representation, likely due to differences in pretraining data and fine-tuning methods.

Prompt sensitivity presents an additional challenge. Minor variations in prompt wording or temperature settings led to measurable changes in bias profiles, suggesting that LLM fairness is not intrinsic but prompt-dependent. Furthermore, while AI paraphrasing reduces bias through lexical simplification, it sometimes erodes the expressive and narrative quality of the text, which is a critical component of educational engagement.

Finally, ethical considerations surrounding the use of AI-generated educational content remain significant. Without transparent disclosure of data provenance and bias evaluation methods, there is a risk of inadvertently amplifying or masking discrimination. Therefore, the results underscore the need for comprehensive frameworks that combine computational fairness assessment with pedagogical and ethical review.

VIII. CONCLUSION AND FUTURE WORK

This research presents a systematic evaluation of bias in educational materials by comparing human-authored NCERT textbook passages with AI-generated paraphrases. The integration of automated measures such as the Word Embedding Association Test (WEAT) and qualitative human annotation enables a comprehensive analysis of both explicit and implicit biases. The results demonstrate that large language models substantially reduce overt gender and

social biases but continue to encode subtle, structural stereotypes learned from large-scale text corpora.

The study establishes that LLMs mitigate explicit bias by approximately 40–50% relative to human-authored materials, primarily by balancing gendered pronoun distribution and neutralizing sentiment polarity. However, implicit stereotypes persist, particularly in contexts involving occupations, communal references, and emotional tone. These findings highlight the potential of LLMs as supportive tools for generating inclusive educational content when used with deliberate prompt design and ongoing human supervision.

Ultimately, the findings contribute to the broader discourse on responsible AI integration in education, emphasizing that equitable learning environments require both algorithmic fairness and human ethical judgment.

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